

Copper layer roster

#	Layer	Role	Function
L1	F.Cu	Signal	100 / 50 Ω · 4 oz · OSFP-XD bus, ELSFP, mgmt I/O
L2	In1.Cu	Reference	Solid GND plane (no splits)
L3	In2.Cu	Signal	HBM4 strobe + DQ stripline 100 Ω diff
L4	In3.Cu	Reference	Solid GND plane
L5	In4.Cu	Power	V_ASIC_0V85 — 24-phase PDN plane
L6	In5.Cu	Reference	Solid GND plane
L7	In6.Cu	Signal	CPO digital lanes · co-packaged interface
L8	In7.Cu	Reference	Solid GND plane
L9	In8.Cu	Reference	GND core reference (centre)
L10	In9.Cu	Reference	Solid GND plane
L11	In10.Cu	Power	V_OPT_1V2 plane · CPO laser bias
L12	In11.Cu	Power	V_AUX_3V3 plane · mgmt rails
L13	In12.Cu	Reference	Solid GND plane
L14	In13.Cu	Signal	PCIe Gen 6 ×16 · 100 Ω diff · back-drilled
L15	In14.Cu	Reference	Solid GND plane
L16	In15.Cu	Signal	Mgmt + OOB · I ² C / SMBus / SPI
L17	In16.Cu	Reference	Solid GND plane
L18	B.Cu	Signal	DrMOS B-side phases + decoupling

Fab output index

Path	Contents
gerbers/	RS-274X · all 27 image layers
drill/	Excellon 2.0 PTH + NPTH + map
ipc2581/	IPC-2581 Rev C unified package
3d/	STEP AP242 board model
native/	KiCad 8.0.9 source files
drawings/	Fab + Assembly + Layer Index PDF
BOM.csv	Manufacturing BOM
*_pnp.csv	Pick-and-place (top + bottom)
DRC_report.rpt	DRC closure report
FAB_READINESS_SIGNOFF.pdf	Investor sign-off
Design_Package.pdf	Architecture + SI/PI/Thermal

Drawing-set legend

Fab Drawing	Mechanical outline · dimensions · drill chart · stackup · 18 fab notes (IPC-6012 Class 3)
Assembly Drawing Top	F-side component placement · 197 footprints · 15 assembly notes · Class 3
Assembly Drawing Bot	B-side component placement (DrMOS phases + decoupling caps)
Layer Index	18-layer copper roster · fab output index · drawing-set legend (this page)
Per-layer PDFs	27 individual layer rasters (gerbers/F.Cu, B.Cu, In1...In16, masks, silks, paste)
Native CAD	LR-S1A_Rev1.0.kicad_pcb — KiCad 8.0.9 schema 20240108

LightRail AI — Confidential / Investor data-room	
Programme	LR-S1A 1U CPO Switch Motherboard
Drawing No.	LR-S1A-IDX-001
Title	Layer Stack-up & Fab Output Index
Material	18-layer HDI · Megtron-7 / FR-4 / BC24
Designer	LightRail AI — Design Automation
Date / Rev	2026-04-19 · Rev 1.01 / 1